The Krasnosel'skiĭ-Mann Iterative Method

Chapter 2: Notation and Mathematical Foundations

Based on Dong, Cho, He, Pardalos & Rassias

SpringerBriefs in Optimization, 2022

Chapter Overview

1. Overview and the Running Example
2. Part (A): Notation
3. Part (B): Classes of Operators
4. Part (C): Convex Analysis
5. Chapter Summary

Outline

1. Overview and the Running Example
2. Part (A): Notation
3. Part (B): Classes of Operators
4. Part (C): Convex Analysis
5. Chapter Summary

Why This Chapter Matters I

Every later chapter in this book studies iterative algorithms of the form $x_{n+1} = T(x_n)$ or blends of such maps, and every convergence proof leans on a small, fixed vocabulary of operator properties (nonexpansive, averaged, monotone, ...) and a handful of standard inequalities. This chapter *is* that vocabulary. Nothing here is specific to the Krasnosel'skiĭ–Mann (KM) method yet — instead we build the toolbox that every subsequent chapter's proofs will reach into without re-explaining.

How to Read This Chapter

We will introduce each definition in plain English with an analogy first, then give the exact formula from the book, then check the definition against a single concrete, running numerical example so that the abstract symbols always have a "home" you can compute by hand.

Why This Chapter Matters II

The Running Example (used throughout this deck)

Let $\mathcal{H} = \mathbb{R}^2$ with the usual dot product, and let

$$C = \{\, z \in \mathbb{R}^2 : \|z\| \le 1 \,\}$$

be the **closed unit disk**. Let $P_C : \mathbb{R}^2 \to C$ be the **metric projection** onto C (formally defined later in this chapter): $P_C(x)$ is the point of C closest to x. For x outside the disk this is $P_C(x) = x/\|x\|$ (rescale x down to the unit circle); for x already inside C, $P_C(x) = x$. We will repeatedly use the two points

$$x = (3, 4), \qquad y = (0, 2).$$

Since $\|x\| = 5 > 1$, $P_C(x) = x/5 = (0.6,\, 0.8)$. Since $\|y\| = 2 > 1$, $P_C(y) = y/2 = (0,\, 1)$. Every operator class introduced below will be checked against this pair.

Outline

1. Overview and the Running Example
2. Part (A): Notation
3. Part (B): Classes of Operators
4. Part (C): Convex Analysis
5. Chapter Summary

Notation You'll Need: Basic Objects I

Spaces and the Ambient Setting

Throughout the book, $\mathbb{N}$ denotes the set of nonnegative integers, and $\mathcal{H}$ and $\mathcal{G}$ denote real **Hilbert spaces** with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. A Hilbert space is simply a vector space equipped with a notion of angle/inner product $\langle a, b \rangle$ (from which the length $\|a\| = \sqrt{\langle a, a \rangle}$ and the distance $\|a - b\|$ are built) that is "complete" (no missing limit points). If you have never seen the general definition, it is safe to think of $\mathcal{H}$ as a (possibly infinite-dimensional) generalization of $\mathbb{R}^N$, the standard N-dimensional Euclidean space, which is itself always a valid (finite-dimensional) Hilbert space. **Every concrete example and Python computation in this deck lives in $\mathcal{H} = \mathbb{R}^2$**, the plane, so you can always picture points as arrows from the origin.

Notation You'll Need: Basic Objects II

Weak Convergence $\rightharpoonup$ versus Strong Convergence $\to$

We write $x_n \to x$ for **strong (ordinary) convergence**: the distance $\|x_n - x\| \to 0$ as $n \to \infty$. We write $x_n \rightharpoonup x$ for **weak convergence**: $\langle x_n, v\rangle \to \langle x, v\rangle$ for *every fixed* $v \in \mathcal{H}$, even though the points x_n themselves need not get uniformly close to x in distance.

Intuition. Strong convergence says "the points really do get close, measured directly." Weak convergence only says "every fixed measurement (projection onto a direction v) of x_n eventually agrees with the same measurement of x." In finite dimensions ($\mathbb{R}^N$, which is where every example in this deck lives), **these two notions coincide exactly**: $x_n \to x$ if and only if $x_n \rightharpoonup x$. So in $\mathbb{R}^2$ you may freely read $\rightharpoonup$ as "$\to$."

Notation You'll Need: Basic Objects III

Why the Distinction Matters at All (a Flag for Later)

In *infinite*-dimensional Hilbert spaces, weak and strong convergence genuinely differ, and several later chapters' convergence proofs only obtain the *weaker* conclusion $x_n \rightharpoonup x^*$. The classic example: let $(e_n)_{n\in\mathbb{N}}$ be an orthonormal sequence in ℓ^2 (e.g., e_n has a 1 in coordinate n and 0 elsewhere). Then $e_n \rightharpoonup 0$ (because $\langle e_n, v\rangle \to 0$ for every fixed $v \in \ell^2$, by Bessel's inequality), yet $\|e_n - 0\| = 1$ for *every* n — the sequence never gets close to 0 in norm, so it does *not* converge strongly. Keep this one-sentence example in your back pocket: it is the standing reason later chapters must work harder to upgrade weak convergence results to strong ones.

Notation You'll Need: Basic Objects IV

The Weak ω-Limit Set $\omega_w(x_n)$

For a sequence $\{x_n\}_{n\in\mathbb{N}}$, define

$$\omega_w(x_n) = \{\, x : \exists\, x_{n_j} \rightharpoonup x \,\}.$$

In words: $\omega_w(x_n)$ collects *every* point that is the weak limit of *some* subsequence of $\{x_n\}$ — the set of all "weak cluster points." **Intuition:** if the whole sequence weakly converges to a single point x, then $\omega_w(x_n) = \{x\}$ is a singleton. If the sequence oscillates between two weak limits (like $(-1)^n$ oscillating between -1 and 1), $\omega_w(x_n)$ contains both. A recurring proof strategy in this book is: show every element of $\omega_w(x_n)$ lies in some target set S, and separately show $\lim_n \|x_n - s\|$ exists for every $s \in S$; together these force $x_n \rightharpoonup$ a single point of S (this is Lemma 2.7 below).

Notation You'll Need: Basic Objects V

A Few More Notational Conventions

- ℓ^1_+ denotes the set of **summable sequences** in $[0, +\infty)$: nonnegative real sequences $\{a_n\}$ with $\sum_{n=0}^{\infty} a_n < +\infty$. These show up as error terms in convergence proofs ("as long as the accumulated error is summable, it cannot prevent convergence").
- For $\xi \in \mathbb{R}$, the **positive part** is $\xi^+ = \max\{0, \xi\}$ and the **negative part** is $\xi^- = -\min\{0, \xi\}$. Both are always ≥ 0, and $\xi = \xi^+ - \xi^-$ while $|\xi| = \xi^+ + \xi^-$. Think of ξ^+ as "the amount by which ξ exceeds 0" and ξ^- as "the amount by which ξ falls short of 0."

Python: Weak = Strong Convergence in $\mathbb{R}^N$, and $\omega_w(x_n)$

In finite dimensions, checking weak convergence reduces to checking $\langle x_n - x, v\rangle \to 0$ for a spanning set of test directions v, equivalent to ordinary convergence $\|x_n - x\| \to 0$. The code verifies this equivalence, and separately illustrates $\omega_w(x_n)$ for an oscillating sequence with two cluster points.

```python
import numpy as np
rng = np.random.default_rng(0)

# Part 1: weak convergence == strong convergence in R^2
x_star = np.array([1.0, -2.0])
xs = [x_star + np.array([1.0, 1.0]) / (n + 1) for n in range(1, 2000)]
test_dirs = rng.normal(size=(5, 2))                # 5 fixed test directions v
strong_err = [np.linalg.norm(xn - x_star) for xn in xs]
weak_err = [max(abs(np.dot(xn - x_star, v)) for v in test_dirs) for xn in xs]
print("strong error -> 0? ", strong_err[-1] < 1e-3)
print("weak error   -> 0? ", weak_err[-1] < 1e-3)    # both vanish together

# Part 2: omega_w(x_n) for an oscillating sequence (two cluster points)
a, b = np.array([1.0, 0.0]), np.array([0.0, 1.0])
seq = [a if n % 2 == 0 else b for n in range(20)]
cluster_points = {tuple(a), tuple(b)}
print("omega_w(x_n) =", cluster_points)          # {(1,0), (0,1)}
```

Outline

1. Overview and the Running Example
2. Part (A): Notation
3. Part (B): Classes of Operators
4. Part (C): Convex Analysis
5. Chapter Summary

Notation You'll Need: Lipschitz, Nonexpansive, Contractive I

Definition 2.1 (Lipschitz-Type Operators)

An operator $T : \mathcal{H} \to \mathcal{H}$ is said to be:

(1) L-**Lipschitz continuous** for $L \geq 0$ if

$$\|T(x) - T(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in \mathcal{H}.$$

(2) **nonexpansive** if $L = 1$.

(3) **contractive** if $0 < L < 1$.

Notation You'll Need: Lipschitz, Nonexpansive, Contractive II

Plain-English Intuition

Lipschitz continuity with constant L says: "T can stretch distances by at most a factor of L." **Nonexpansive** $(L = 1)$ means T *never stretches distances at all* — two points can only get closer together or stay the same distance apart after applying T, never farther apart. **Contractive** $(L < 1)$ is stronger still: T must *strictly shrink* every pair of distances by a fixed factor, which is exactly the hypothesis of the Banach fixed-point theorem.

Notation You'll Need: Lipschitz, Nonexpansive, Contractive III

Running Example: P_C is Nonexpansive

Using $x = (3, 4)$, $y = (0, 2)$ from before, with $P_C(x) = (0.6, 0.8)$ and $P_C(y) = (0, 1)$:

$$\|P_C(x) - P_C(y)\| = \|(0.6,\ -0.2)\| = \sqrt{0.36 + 0.04} = \sqrt{0.4} \approx 0.632,$$

$$\|x - y\| = \|(3, 2)\| = \sqrt{9 + 4} = \sqrt{13} \approx 3.606.$$

Indeed $0.632 \leq 3.606$: projecting onto C did not increase the distance between the two points. (We verify this holds for *every* pair x, y, not just this one, in Lemma 2.4 below.)

Notation You'll Need: Lipschitz, Nonexpansive, Contractive IV

Definition 2.2 (Pseudocontractive Operators)

An operator $T : \mathcal{H} \to \mathcal{H}$ is **pseudocontractive** if

$$\|T(x) - T(y)\|^2 \leq \|x - y\|^2 + \|(\mathrm{Id} - T)x - (\mathrm{Id} - T)y\|^2 \quad \text{for all } x, y \in \mathcal{H}.$$

Here Id is the **identity operator**, $\mathrm{Id}(x) = x$. The class of pseudocontractive operators *includes* the nonexpansive operators: if T is nonexpansive, the extra term $\|(\mathrm{Id} - T)x - (\mathrm{Id} - T)y\|^2 \geq 0$ only makes the right-hand side larger, so the inequality
$\|T(x) - T(y)\|^2 \leq \|x - y\|^2 \leq \|x - y\|^2 + \|(\mathrm{Id} - T)x - (\mathrm{Id} - T)y\|^2$ holds automatically.
Pseudocontractive operators are thus a strictly larger, more permissive class.

Notation You'll Need: Demicontractive and Quasi-nonexpansive Operators I

Definition 2.3 (k-Demicontractive Operators)

An operator $T : \mathcal{H} \to \mathcal{H}$ **with a fixed point** (recall $\mathrm{Fix}(T) = \{x \in \mathcal{H} : T(x) = x\}$, the set of points left unchanged by T) is said to be:

(1) k**-demicontractive** for $k \in \mathbb{R}$ if

$$\|T(x) - y\|^2 \leq \|x - y\|^2 + k\|T(x) - x\|^2 \quad \text{for all } x \in \mathcal{H},\ y \in \mathrm{Fix}(T).$$

(2) **quasi-nonexpansive** if $k = 0$, i.e. $\|T(x) - y\| \leq \|x - y\|$ for all $x \in \mathcal{H}$, $y \in \mathrm{Fix}(T)$.

Notation You'll Need: Demicontractive and Quasi-nonexpansive Operators II

Plain-English Intuition

These definitions only constrain how T behaves *relative to its own fixed points* $y \in \mathrm{Fix}(T)$, not relative to arbitrary pairs of points. Quasi-nonexpansive says: "every point moves strictly closer to (or no farther from) every fixed point after applying T." This is weaker than nonexpansiveness, which demands the distance-preserving property for *all* pairs $x, y \in \mathcal{H}$, fixed points or not. **If T has a fixed point and is nonexpansive, then T is automatically quasi-nonexpansive** (just specialize y to be a fixed point in the nonexpansive inequality), **but the reverse is not necessarily true** — there are quasi-nonexpansive maps that stretch distances between two arbitrary (non-fixed) points.

Notation You'll Need: Demicontractive and Quasi-nonexpansive Operators III

Running Example: P_C is Quasi-nonexpansive

$\mathrm{Fix}(P_C) = C$ itself: every point already in the unit disk is left unchanged, and no point outside C is fixed. Since P_C is nonexpansive (previous frame) and has fixed points, it is automatically quasi-nonexpansive: for any $x \in \mathbb{R}^2$ and any $y \in C$, $\|P_C(x) - y\| \leq \|x - y\|$. Take $x = (3, 4)$ and the fixed point $y = (0.6, 0.8) \in C$ (an arbitrary boundary point of C):

$$\|P_C(x) - y\| = \|(0.6, 0.8) - (0.6, 0.8)\| = 0 \leq \|x - y\| = \|(2.4, 3.2)\| = 4.$$

Firmly Nonexpansive Operators I

Definition 2.4 (Firmly Nonexpansive)

An operator $T : \mathcal{H} \to \mathcal{H}$ is said to be **firmly nonexpansive** if

$$\|T(x) - T(y)\|^2 \leq \langle T(x) - T(y),\, x - y \rangle \quad \text{for all } x, y \in \mathcal{H}.$$

Firmly Nonexpansive Operators II

Plain-English Intuition

Firm nonexpansiveness is a *stronger*, better-behaved kind of nonexpansiveness. Geometrically, it says the vector $T(x) - T(y)$ and the vector $x - y$ point in a sufficiently "aligned" direction that their inner product dominates the squared length of $T(x) - T(y)$ — think of it as "the image points can never overshoot, and in fact the angle between the displacement of the inputs and the displacement of the outputs is never obtuse." **If T is firmly nonexpansive, then T is nonexpansive**: by the Cauchy–Schwarz inequality,

$$\|T(x) - T(y)\|^2 \leq \langle T(x) - T(y), x - y \rangle \leq \|T(x) - T(y)\| \, \|x - y\|,$$

and dividing both sides by $\|T(x) - T(y)\|$ (when nonzero) gives $\|T(x) - T(y)\| \leq \|x - y\|$.

Firmly Nonexpansive Operators III

Lemma 2.1 (Equivalent Characterization)

Let $T : \mathcal{H} \to \mathcal{H}$. The following are equivalent:

(1) T is firmly nonexpansive.

(2) $2T - \mathrm{Id}$ is nonexpansive.

Step-by-Step Verification for P_C (Lemma 2.4(1))

We verify that P_C is firmly nonexpansive using the running example $x = (3, 4)$, $y = (0, 2)$, $P_C(x) = (0.6, 0.8)$, $P_C(y) = (0, 1)$:

$$\text{LHS} = \|P_C(x) - P_C(y)\|^2 = \|(0.6, -0.2)\|^2 = 0.36 + 0.04 = 0.4,$$

$$\text{RHS} = \langle P_C(x) - P_C(y),\, x - y\rangle = \langle (0.6, -0.2), (3, 2)\rangle = 1.8 - 0.4 = 1.4.$$

Indeed $0.4 \leq 1.4$, confirming the firmly-nonexpansive inequality for this pair. (Below we derive this fact for *all* x, y, using the variational characterization of the projection.)

Firmly Nonexpansive Operators IV

Definition 2.5 (α-Averaged Operators)

Let $T : \mathcal{H} \to \mathcal{H}$. For $\alpha \in (0, 1)$, T is said to be α-**averaged** if there exists a nonexpansive operator $S : \mathcal{H} \to \mathcal{H}$ such that

$$T = \alpha S + (1 - \alpha)\mathrm{Id}.$$

Plain-English Intuition

An averaged operator is a **weighted blend of the identity map and a nonexpansive map**: applying T means moving only a fraction α of the way from "do nothing" (Id) toward "apply the nonexpansive map S." This "partial step" structure is *literally what powers the Krasnosel'skiĭ–Mann iteration*: $x_{n+1} = (1 - \alpha_n)x_n + \alpha_n T(x_n)$ is precisely an averaging step, and understanding when the resulting map is averaged (hence well-behaved) is central to the rest of this book.

Firmly Nonexpansive Operators V

Remark 2.1

If T is firmly nonexpansive, then T is $\frac{1}{2}$-averaged. (Proof: by Lemma 2.1, $S := 2T - \mathrm{Id}$ is nonexpansive, and rearranging gives exactly $T = \frac{1}{2}S + \frac{1}{2}\mathrm{Id}$.)

Firmly Nonexpansive Operators VI

Running Example: P_C Is $\frac{1}{2}$-Averaged

Since P_C is firmly nonexpansive, Remark 2.1 gives $P_C = \frac{1}{2}S + \frac{1}{2}\mathrm{Id}$ with $S = 2P_C - \mathrm{Id}$ (this S is called the **reflection** of P_C, see Definition 2.6 below). Check S is nonexpansive numerically at our running pair: $S(x) = 2(0.6, 0.8) - (3, 4) = (-1.8, -2.4)$ and $S(y) = 2(0, 1) - (0, 2) = (0, 0)$, so

$$\|S(x) - S(y)\| = \|(-1.8, -2.4)\| = \sqrt{3.24 + 5.76} = \sqrt{9} = 3, \qquad \|x - y\| = \sqrt{13} \approx 3.606,$$

and $3 \leq 3.606$: S does not stretch this pair, consistent with S being nonexpansive.

Lemma 2.2 (Averagedness Constant of a Composition)

Let C be a nonempty subset of $\mathcal{H}$, $T_1 : C \to C$ be σ_1-averaged, and $T_2 : C \to C$ be σ_2-averaged for $\sigma_1, \sigma_2 \in (0,1)$. Set $T = T_1 T_2$ (composition) and

$$\sigma = \frac{\sigma_1 + \sigma_2 - 2\sigma_1\sigma_2}{1 - \sigma_1\sigma_2}. \tag{2.1}$$

Then $\sigma \in (0,1)$ and T is σ-averaged.

Remark 2.2 (Other Averagedness Constants) and a Worked Example

Besides (2.1), two other constants also make the composition averaged:

$$\sigma = \frac{2\max\{\sigma_1, \sigma_2\}}{\max\{\sigma_1, \sigma_2\} + 1} \quad \text{and} \quad \sigma = \sigma_1 + \sigma_2 - \sigma_1\sigma_2.$$

Combettes and Yamada compared these three constants and showed that (2.1) is always the smallest (best); Huang et al. later proved (2.1) is tight (cannot be improved in general).

Worked example. Take $T_1 = T_2 = P_C$, so $\sigma_1 = \sigma_2 = \frac{1}{2}$. Formula (2.1) gives

$$\sigma = \frac{\frac{1}{2} + \frac{1}{2} - 2 \cdot \frac{1}{4}}{1 - \frac{1}{4}} = \frac{1 - 0.5}{0.75} = \frac{0.5}{0.75} = \frac{2}{3}.$$

So Lemma 2.2 certifies $T = P_C \circ P_C$ is $\frac{2}{3}$-averaged. Since $P_C \circ P_C = P_C$ (projecting twice does nothing new), and we already know P_C is $\frac{1}{2}$-averaged, this shows the composition formula gives a *valid but not always the tightest possible* constant for special (e.g. identical) operators — $\frac{1}{2} < \frac{2}{3}$, and both certify averagedness.

Relaxation of an Operator I

Definition 2.6 (Relaxation)

Let $S : \mathcal{H} \to \mathcal{H}$ and $\alpha \in (0, +\infty)$. The operator $T = \mathrm{Id} + \alpha(S - \mathrm{Id})$ is called a **relaxation** of S. Furthermore:

(1) If $\alpha \leq 1$, T is an **underrelaxation** of S.

(2) If $\alpha > 1$, T is an **overrelaxation** of S.

(3) If $\alpha = 2$, T is the **reflection** of S (matches the $S = 2P_C - \mathrm{Id}$ used above, with $S \to P_C$ playing the role of the base operator being reflected).

Connecting Definitions 2.5 and 2.6

Expanding $\mathrm{Id} + \alpha(S - \mathrm{Id}) = \alpha S + (1 - \alpha)\mathrm{Id}$ shows a relaxation with parameter α is *literally* the same formula as an α-averaging of S. Consequently: **an averaged operator is exactly an underrelaxation ($\alpha \in (0, 1)$) of one nonexpansive operator.**

The Metric Projection P_C: Definition and Variational Characterization I

Definition 2.9 (Metric Projection)

Let C be a nonempty closed and convex subset of $\mathcal{H}$. For each point $x \in \mathcal{H}$, there exists a **unique** nearest point in C, denoted $P_C(x)$, that is,

$$\|x - P_C(x)\| \leq \|x - y\| \quad \text{for all } y \in C. \tag{2.2}$$

The mapping $P_C : \mathcal{H} \to C$ is called the **metric projection** of $\mathcal{H}$ onto C.

Intuition

$P_C(x)$ answers: "of all the points allowed to be in C, which one is closest to x?" Existence and uniqueness of this nearest point is a basic (but nontrivial) consequence of C being closed and convex in a Hilbert space — convexity rules out two equally-close points on "opposite sides" of a non-convex set.

The Metric Projection P_C: Definition and Variational Characterization II

Lemma 2.3 (Variational Characterization of P_C)

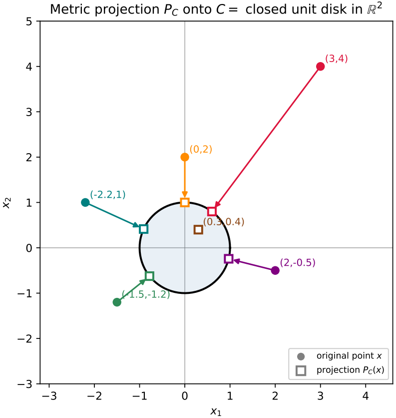
Let $x \in \mathcal{H}$ and $z \in C$. Then $z = P_C(x)$ **if and only if**

$$\langle x - z,\, z - y \rangle \geq 0 \quad \text{for all } y \in C. \tag{2.3}$$

Intuition for (2.3)

This says: standing at the projected point z, the direction back toward x (namely $x - z$) makes a *non-obtuse* angle with the direction toward any other point $y \in C$ (namely $z - y$). Geometrically, the whole set C lies in the half-space "behind" the hyperplane through z perpendicular to $x - z$ — exactly the picture of a nearest-point projection onto a convex set.

The Metric Projection P_C: Definition and Variational Characterization III



Lemma 2.4: The Key Projection Inequalities I

Lemma 2.4

For any $x, y \in \mathcal{H}$ and $z \in C$, the following hold:

(1) $\|P_C(x) - P_C(y)\|^2 \leq \langle P_C(x) - P_C(y),\, x - y\rangle$ (P_C *is firmly nonexpansive*).

(2) $\|P_C(x) - z\|^2 \leq \|x - z\|^2 - \|P_C(x) - x\|^2$ (*a Pythagorean-type shrinkage inequality*).

(3) $\langle (\mathrm{Id} - P_C)x - (\mathrm{Id} - P_C)y,\, x - y\rangle \geq \|(\mathrm{Id} - P_C)x - (\mathrm{Id} - P_C)y\|^2$ ($\mathrm{Id} - P_C$ *is itself firmly nonexpansive*).

Lemma 2.4: The Key Projection Inequalities II

Derivation of (1) from the Variational Characterization (2.3)

Let $u = P_C(x)$, $v = P_C(y)$. Applying (2.3) with $z = u$, $y \to v$ (as the free point of C) gives $\langle x - u, u - v\rangle \geq 0$. Applying (2.3) with $z = v$, $y \to u$ gives $\langle y - v, v - u\rangle \geq 0$, i.e. $\langle y - v, u - v\rangle \leq 0$. Subtracting,

$$\langle x - u, u - v\rangle - \langle y - v, u - v\rangle \geq 0 \implies \langle (x - u) - (y - v),\, u - v\rangle \geq 0 \implies \langle x - y,\, u - v\rangle \geq \|u - v\|^2,$$

which is exactly $\langle P_C(x) - P_C(y), x - y\rangle \geq \|P_C(x) - P_C(y)\|^2$, i.e. statement (1).

Lemma 2.4: The Key Projection Inequalities III

Numeric Check of (1), Continued from Before

We already verified for $x = (3, 4)$, $y = (0, 2)$: LHS $= 0.4$, RHS $= 1.4$, so $0.4 \le 1.4$ holds. This numeric check, plus the general derivation above, together confirm (1) both abstractly and concretely.

Derivation that $\mathrm{Id} - P_C$ Is $\frac{1}{2}$-Cocoercive

(Cocoercivity is defined precisely in Part (C); for now, B is β-cocoercive if $\beta\|Bx - By\|^2 \le \langle Bx - By, x - y\rangle$ for all x, y.) We show: **if T is nonexpansive** (as P_C is), then $B := \mathrm{Id} - T$ is $\frac{1}{2}$-cocoercive. Write $u = x - y$ and $v = T(x) - T(y)$, so $Bx - By = u - v$. Then

$$\langle u - v,\, u\rangle - \tfrac{1}{2}\|u - v\|^2 = \left(\|u\|^2 - \langle v, u\rangle\right) - \tfrac{1}{2}\left(\|u\|^2 - 2\langle u, v\rangle + \|v\|^2\right) = \tfrac{1}{2}\left(\|u\|^2 - \|v\|^2\right).$$

Since T is nonexpansive, $\|v\| = \|T(x) - T(y)\| \le \|x - y\| = \|u\|$, so $\|u\|^2 - \|v\|^2 \ge 0$. Hence $\langle(\mathrm{Id} - T)x - (\mathrm{Id} - T)y,\, x - y\rangle \ge \frac{1}{2}\|(\mathrm{Id} - T)x - (\mathrm{Id} - T)y\|^2$, which is exactly $\frac{1}{2}$-cocoercivity of $\mathrm{Id} - T$.

Lemma 2.4: The Key Projection Inequalities IV

Numeric Confirmation for the Running Example

With $u = x - y = (3, 2)$ and $v = P_C(x) - P_C(y) = (0.6, -0.2)$: $\|u\|^2 = 13$, $\|v\|^2 = 0.4$. Then $(\mathrm{Id} - P_C)x - (\mathrm{Id} - P_C)y = u - v = (2.4, 2.2)$, so

$$\langle u - v, u\rangle = \langle(2.4, 2.2), (3, 2)\rangle = 7.2 + 4.4 = 11.6, \qquad \tfrac{1}{2}\|u - v\|^2 = \tfrac{1}{2}(5.76 + 4.84) = \tfrac{1}{2}(10.6) = 5.3.$$

Indeed $11.6 \geq 5.3$, confirming $\frac{1}{2}$-cocoercivity numerically. (In fact Lemma 2.4(3) gives the *sharper* inequality $11.6 \geq 10.6$ – i.e. $\mathrm{Id} - P_C$ is even firmly nonexpansive, a stronger property than mere $\frac{1}{2}$-cocoercivity; general nonexpansive T only guarantees the weaker $\frac{1}{2}$-cocoercive bound derived above.)

Fixed-Point Sets, Demiclosedness, and Two Workhorse Lemmas I

Lemma 2.5 ($\mathrm{Fix}(T)$ Is Closed and Convex)

Let $T : \mathcal{H} \to \mathcal{H}$ be nonexpansive. Then $\mathrm{Fix}(T)$ is closed and convex.

Why This Matters

Knowing $\mathrm{Fix}(T)$ is closed and convex means it behaves like a "nice" target set: it has well-defined projections onto it, and limits of sequences in $\mathrm{Fix}(T)$ stay in $\mathrm{Fix}(T)$. For our running example, $\mathrm{Fix}(P_C) = C$, the closed unit disk, which is indeed closed and convex.

Lemma 2.6 (Demiclosedness Principle Consequence)

Let $C \subseteq \mathcal{H}$ be nonempty closed and convex and $T : C \to \mathcal{H}$ be a nonexpansive mapping. Let $\{x_n\}$ be a sequence in C and let $x \in \mathcal{H}$ be such that $x_n \rightharpoonup x$ and $T(x_n) - x_n \to 0$ as $n \to +\infty$. Then $x \in \mathrm{Fix}(T)$.

Fixed-Point Sets, Demiclosedness, and Two Workhorse Lemmas II

Plain-English Intuition

"Demiclosed" means: if a sequence converges weakly *and* its residual $T(x_n) - x_n$ (how far each point is from being fixed) shrinks to zero *strongly*, then the weak limit really is an honest fixed point of T – the map $\mathrm{Id} - T$ cannot "hide" a discrepancy that vanishes in residual but not in the limit itself. This is the standard tool used to show that the limit produced by an iterative algorithm is actually a solution (a fixed point), not just a point where the algorithm happens to slow down.

Fixed-Point Sets, Demiclosedness, and Two Workhorse Lemmas III

Lemma 2.7 (Weak Convergence Criterion – an Opial-Type Lemma)

Let C be a nonempty subset of $\mathcal{H}$ and $\{x_n\}$ a sequence in $\mathcal{H}$ such that:

(a) for all $x \in C$, $\lim_{n\to\infty} \|x_n - x\|$ exists;

(b) every sequential weak cluster point of $\{x_n\}$ (i.e. every element of $\omega_w(x_n)$) is in C.

Then the sequence $\{x_n\}$ converges weakly to a point in C.

Why This Matters

This is the standard two-step recipe used throughout the book to prove weak convergence of an iterative method: (a) show the distance from x_n to *every* candidate solution shrinks monotonically (so a limit exists), and (b) show that any weak cluster point must be a genuine solution (using, e.g., Lemma 2.6). Together, these facts upgrade “the sequence doesn't diverge” into “the sequence weakly converges to a single specific solution.”

Lemma 2.8 (A Discrete Gronwall / Almost-Supermartingale Lemma)

Assume $\{a_n\}$ is a sequence of nonnegative real numbers such that

$$a_{n+1} \le (1+\gamma_n)a_n + \delta_n \quad \text{for each } n \ge 0,$$

where $\{\gamma_n\} \subset [0, +\infty)$ and $\{\delta_n\}$ satisfy:

(a) $\sum_{n=0}^{\infty} \gamma_n < +\infty$;

(b) $\sum_{n=0}^{\infty} \delta_n < +\infty$ or $\limsup_{n\to\infty} \delta_n \le 0$.

Then $\lim_{n\to\infty} a_n$ exists.

Fixed-Point Sets, Demiclosedness, and Two Workhorse Lemmas V

Intuition

This is a quantitative version of "if the growth factors and the injected errors are both summable (or eventually non-positive), the sequence cannot blow up and in fact settles down." It is the standard engine for proving that error terms $a_n = \|x_n - x^*\|^2$ in perturbed / inertial iterative methods still converge, even though the exact monotone-decrease property may fail at each individual step.

Fixed-Point Sets, Demiclosedness, and Two Workhorse Lemmas VI

Lemma 2.9 and a Useful Algebraic Identity

For any $a, b \in \mathcal{H}$,

$$\|a-b\|^2 \leq (1+\|b\|)\|a\|^2 + \|b\| + \|b\|^2.$$

This bound (provable via Cauchy–Schwarz and the mean value inequality) lets you control $\|a-b\|^2$ using only $\|a\|^2$ and $\|b\|$, which is convenient when b is a small perturbation term.

The following identity is used repeatedly in later chapters: for all $\alpha \in \mathbb{R}$ and $(x, y) \in \mathcal{H} \times \mathcal{H}$,

$$\|\alpha x + (1-\alpha)y\|^2 = \alpha\|x\|^2 + (1-\alpha)\|y\|^2 - \alpha(1-\alpha)\|x-y\|^2. \tag{2.4}$$

Intuition for (2.4): the squared norm of a weighted average of x and y equals the weighted average of the squared norms, *minus* a correction term that grows with how far apart x and y are. This correction term is exactly what makes averaged/KM-type updates provably contract distances to a fixed point – it is the algebraic heart of nearly every convergence proof in this book.

Python: Nonexpansive, Firmly Nonexpansive, and Averaged — Numeric Check

The code below checks, for many random pairs of points in $\mathbb{R}^2$, that P_C (projection onto the closed unit disk) is simultaneously nonexpansive, firmly nonexpansive, and that its reflection $S = 2P_C - \mathrm{Id}$ is nonexpansive (confirming P_C is $\frac{1}{2}$-averaged via Remark 2.1).

```python
import numpy as np
rng = np.random.default_rng(1)

def P_C(z):                          # projection onto closed unit disk
    n = np.linalg.norm(z)
    return z if n <= 1.0 else z / n

ne_ok, fne_ok, avg_ok = True, True, True
for _ in range(5000):
    x, y = rng.uniform(-5, 5, 2), rng.uniform(-5, 5, 2)
    px, py = P_C(x), P_C(y)
    dp, dxy = px - py, x - y
    if np.linalg.norm(dp) > np.linalg.norm(dxy) + 1e-9:
        ne_ok = False                           # (1) nonexpansive
    if np.dot(dp, dp) > np.dot(dp, dxy) + 1e-9:
        fne_ok = False                          # (2) firmly nonexpansive
    Sx, Sy = 2 * px - x, 2 * py - y             # (3) reflection S=2P_C-Id
    if np.linalg.norm(Sx - Sy) > np.linalg.norm(dxy) + 1e-9:
        avg_ok = False                          #     is nonexpansive?

print("nonexpansive?          ", ne_ok)
```

Outline

1. Overview and the Running Example
2. Part (A): Notation
3. Part (B): Classes of Operators
4. Part (C): Convex Analysis
5. Chapter Summary

Notation You'll Need: Set-Valued Operators and Monotonicity I

Set-Valued Operators: Domain, Range, Graph, Inverse, Zeros

Let $A : \mathcal{H} \rightrightarrows \mathcal{H}$ be a **set-valued operator**: for each $x \in \mathcal{H}$, Ax is a (possibly empty) *set* of points in $\mathcal{H}$, rather than a single point. Think of an ordinary function as a special case where every Ax has exactly one element.

(1) The **domain** of A is $\operatorname{dom} A = \{x \in \mathcal{H} : Ax \neq \emptyset\}$.

(2) The **range** of A is $\operatorname{ran} A = \{y \in \mathcal{H} : \exists x \in \mathcal{H},\ y \in Ax\}$.

(3) The **graph** of A is $\operatorname{gra} A = \{(x, y) \in \mathcal{H}^2 : y \in Ax\}$.

(4) The **inverse** of A is the operator whose graph is $\operatorname{gra} A^{-1} = \{(y, x) \in \mathcal{H}^2 : x \in A^{-1}y\}$.

(5) The set of **zeros** is $\operatorname{zer} A = \{x \in \mathcal{H} : 0 \in Ax\} = A^{-1}(0)$.

Notation You'll Need: Set-Valued Operators and Monotonicity II

Intuition

Set-valued operators generalize functions to allow, e.g., "kinks" (like the subdifferential of $|x|$ at $x = 0$, which is the whole interval $[-1, 1]$ rather than a single slope). $\operatorname{zer} A$ is exactly the solution set of the inclusion problem "find x with $0 \in Ax$," which is the abstract template underlying root-finding, optimization (via Fermat's theorem below), and variational inequalities all at once.

Notation You'll Need: Set-Valued Operators and Monotonicity III

Definition 2.10 (Monotone and Maximally Monotone Operators)

(1) A set-valued operator $A : \mathcal{H} \rightrightarrows \mathcal{H}$ is **monotone** if

$$\langle x - y,\, u - v \rangle \geq 0 \quad \text{for all } (x, u), (y, v) \in \operatorname{gra} A.$$

(2) A is **maximally monotone** if $\operatorname{gra} A$ is not strictly contained in the graph of any other monotone operator (i.e. it is a monotone operator that cannot be "enlarged" while staying monotone).

Notation You'll Need: Set-Valued Operators and Monotonicity IV

Plain-English Intuition

Monotonicity is the **multivariate generalization of "increasing function."** In one dimension, $a : \mathbb{R} \to \mathbb{R}$ is a nondecreasing function exactly when $(x - y)(a(x) - a(y)) \geq 0$ for all x, y – moving x up never decreases $a(x)$ relative to $a(y)$. The inner-product version $\langle x - y, u - v \rangle \geq 0$ says the same thing for multi-dimensional / set-valued maps: displacement in the input and displacement in the output never point in "opposite" directions. Maximality is a technical completeness condition ensuring the operator is not missing any pairs it could consistently include – it is what makes resolvents (defined next) well-defined everywhere.

Notation You'll Need: Set-Valued Operators and Monotonicity V

Definition 2.11 (Resolvent, Reflection Resolvent, Yosida Approximation)

Let $A : \mathcal{H} \rightrightarrows \mathcal{H}$ be maximally monotone.

(1) For $\gamma > 0$, the **resolvent** of A is
$$J_{\gamma A} = (\mathrm{Id} + \gamma A)^{-1}. \tag{2.5}$$

(2) For $\gamma > 0$, the **reflection resolvent** is
$$R_{\gamma A} = 2J_{\gamma A} - \mathrm{Id}. \tag{2.6}$$

(3) For $\gamma \in \mathbb{R}$, the **Yosida approximation** of A with parameter γ is
$$A_\gamma = (A^{-1} + \gamma \mathrm{Id})^{-1}.$$

Notation You'll Need: Set-Valued Operators and Monotonicity VI

Intuition

The resolvent $J_{\gamma A}$ turns the (possibly set-valued, hard-to-use) operator A into a genuine, single-valued, well-behaved map – this is exactly how the KM iteration and operator-splitting methods (Chapter 4) apply to inclusions $0 \in Ax$ that are not ordinary equations. The reflection resolvent mirrors the role of the reflection $2T - \mathrm{Id}$ seen earlier for nonexpansive T.

Notation You'll Need: Set-Valued Operators and Monotonicity VII

Lemma 2.10 (Resolvents Are Firmly Nonexpansive, and Conversely)

Let $A : \mathcal{H} \rightrightarrows \mathcal{H}$ be maximally monotone and $T : \mathcal{H} \to \mathcal{H}$ be firmly nonexpansive. Then:

(1) J_A is firmly nonexpansive.

(2) There exists a maximally monotone operator $A' : \mathcal{H} \rightrightarrows \mathcal{H}$ such that $T = J_{A'}$.

An Important Subtlety

By Lemma 2.1(2) and Lemma 2.10(1), the reflection resolvent $R_{\gamma A}$ is *nonexpansive*. **However, it is not firmly nonexpansive or averaged in general** – reflections are a genuinely different, weaker class of well-behaved operator, important later for splitting methods (e.g. Douglas–Rachford) that use reflections rather than plain resolvents.

Cococercive Operators I

Definition 2.12 (β-Cocoercive Operators)

An operator $B : \mathcal{H} \to \mathcal{H}$ is said to be β-**cocoercive** for some $\beta \in (0, +\infty)$ if

$$\beta \|Bx - By\|^2 \leq \langle Bx - By,\, x - y \rangle \quad \text{for all } x, y \in \mathcal{H}.$$

Plain-English Intuition

Cocoercivity is a **quantified, stronger form of monotone + Lipschitz combined**: it says not only that B is monotone ($\langle Bx - By, x - y \rangle \geq 0$, taking $\beta > 0$), but that the inner product must be *at least* β times the squared output distance – a precise, checkable trade-off between how monotone and how Lipschitz B is. Observe that β-cocoercivity implies $\frac{1}{\beta}$-Lipschitz continuity (by Cauchy–Schwarz, exactly as in the firmly-nonexpansive argument above).

Cocoercive Operators II

The Key Relation to Nonexpansiveness

If T is nonexpansive, then $\mathrm{Id} - T$ is $\frac{1}{2}$-cocoercive. The converse is also true: if $\mathrm{Id} - T$ is $\frac{1}{2}$-cocoercive then T is nonexpansive. We proved the forward direction explicitly above (Lemma 2.4 discussion), by algebra plus $\|T(x) - T(y)\| \leq \|x - y\|$.

Cocoercive Operators III

Lemma 2.11

Let $B : \mathcal{H} \to \mathcal{H}$ be β-cocoercive for some $\beta > 0$. Then:

(1) βB is firmly nonexpansive.

(2) $\mathrm{Id} - \gamma B$ is $\frac{\gamma}{2\beta}$-averaged for $\gamma \in (0, 2\beta)$.

Running Example: Applying Lemma 2.11 to $B = \mathrm{Id} - P_C$

We showed $B = \mathrm{Id} - P_C$ is $\frac{1}{2}$-cocoercive ($\beta = \frac{1}{2}$). By Lemma 2.11(1), $\beta B = \frac{1}{2}(\mathrm{Id} - P_C)$ is firmly nonexpansive. By Lemma 2.11(2) with $\gamma \in (0, 1)$, $\mathrm{Id} - \gamma(\mathrm{Id} - P_C) = (1 - \gamma)\mathrm{Id} + \gamma P_C$ is $\frac{\gamma}{1} = \gamma$-averaged – exactly recovering that any convex combination of Id and P_C with weight $\gamma \in (0, 1)$ on P_C is γ-averaged, consistent with Definition 2.5 applied directly to $S = P_C$.

Cocoercive Operators IV

Definition 2.13 (Uniformly Monotone Operators)

An operator $A : \mathcal{H} \rightrightarrows \mathcal{H}$ is **uniformly monotone** if there exists an increasing function $\phi_A : [0, +\infty) \to [0, +\infty)$ that vanishes only at 0 and satisfies

$$\langle x - y,\, u - v\rangle \geq \phi_A(\|x - y\|) \quad \text{for all } (x, u), (y, v) \in \operatorname{gra} A.$$

Cocoercive Operators V

Definition 2.14 (Strongly Monotone and Quasi-Strongly Monotone)

Let $\gamma > 0$ be arbitrary.

(1) $A : \mathcal{H} \to \mathcal{H}$ is γ-**strongly monotone** if

$$\langle x - y,\, Ax - Ay\rangle \geq \gamma \|x - y\|^2 \quad \text{for all } x, y \in \mathcal{H}.$$

(2) A is **quasi-γ-strongly monotone** if

$$\langle x - y,\, Ax\rangle \geq \gamma \|x - y\|^2 \quad \text{for all } x \in \mathcal{H},\ y \in \mathrm{zer}(A).$$

A strongly monotone operator is a prominent representative of the uniformly monotone class (take $\phi_A(t) = \gamma t^2$). If A is γ-strongly monotone with $\mathrm{zer}(A) \neq \emptyset$, it is automatically quasi-γ-strongly monotone.

Cocoercive Operators VI

Definition 2.15 (Parallel Sum)

Let $C, D : \mathcal{H} \rightrightarrows \mathcal{H}$ be two set-valued operators. The **parallel sum** of C and D is defined by

$$C \,\square\, D := (C^{-1} + D^{-1})^{-1}.$$

(This mirrors the “resistors in parallel” formula from circuit theory, and shows up in operator-splitting methods combining two monotone operators, Chapter 4.)

Convex Functions, the Subdifferential, and Fermat's Theorem I

Definition 2.16 (Convex, Lower Semi-continuous, Proper Functions)

(1) $f : \mathcal{H} \to (-\infty, +\infty]$ is **convex** if for all $x, y \in \mathcal{H}$ and $\alpha \in (0,1)$,

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y).$$

(2) f is **lower semi-continuous (lsc)** if for every sequence $\{x_n\} \subset \mathcal{H}$ and $x \in \mathcal{H}$,

$$x_n \to x \implies f(x) \leq \varliminf_{n\to\infty} f(x_n).$$

(3) f is **proper** if $\operatorname{dom} f = \{x \in \mathcal{H} : f(x) < +\infty\} \neq \emptyset$.

Denote by $\Gamma_0(\mathcal{H})$ the set of proper lower semi-continuous convex functions from $\mathcal{H}$ to $(-\infty, +\infty]$.

Convex Functions, the Subdifferential, and Fermat's Theorem II

Intuition

Convexity: the graph of f never lies above the straight line segment joining two of its points (f "bows downward"). Allowing values up to $+\infty$ (rather than requiring f finite everywhere) is the standard trick used to encode *constraints* as functions (see the indicator function below). Lower semi-continuity is a mild regularity condition ruling out functions that can suddenly "jump down" at a limit point – it is exactly what is needed for minimizers to exist and for subdifferentials to behave well. $\Gamma_0(\mathcal{H})$ is simply "the well-behaved convex functions we are allowed to optimize."

Convex Functions, the Subdifferential, and Fermat's Theorem III

Definition 2.17 (Subdifferential)

Let $f \in \Gamma_0(\mathcal{H})$.

(1) The **subdifferential** of f is the set-valued operator

$$\partial f(x) := \{\, g \in \mathcal{H} : \langle x' - x,\, g \rangle + f(x) \leq f(x'),\ \forall x' \in \mathcal{H} \,\}.$$

(2) f is **subdifferentiable** at x if $\partial f(x) \neq \emptyset$.

(3) An element of $\partial f(x)$ is called a **subgradient**.

Plain-English Intuition

The subdifferential is **the set of all valid supporting slopes of a possibly-kinked convex function, generalizing the derivative**. A subgradient g at x is any slope such that the tangent line/hyperplane $f(x) + \langle g, \cdot - x \rangle$ stays *below* the graph of f everywhere. Where f is smooth, $\partial f(x) = \{\nabla f(x)\}$ is a single point (the ordinary gradient). Where f has a "kink" (like $|x|$ at 0), $\partial f(x)$ is a whole *set* of valid slopes (for $|x|$ at $x = 0$: $\partial f(0) = [-1, 1]$).

Lemma 2.12

If $f \in \Gamma_0(\mathcal{H})$, then ∂f is maximally monotone. (This links convex analysis directly back to Part (C)'s monotone-operator machinery: minimizing a convex function is the same kind of problem as finding a zero of a maximally monotone operator.)

The Indicator Function and the Normal Cone

Let C be a nonempty closed convex subset of $\mathcal{H}$. The **indicator function** ι_C is defined by

$$\iota_C(x) = \begin{cases} 0, & x \in C, \\ +\infty, & \text{otherwise.} \end{cases}$$

The **normal cone operator** N_C of C is the subdifferential of ι_C, i.e. $N_C = \partial \iota_C$, given by

$$N_C(x) = \begin{cases} \{u \in \mathcal{H} : \langle y - x,\, u \rangle \leq 0,\ \forall y \in C\}, & \text{if } x \in C, \\ \emptyset, & \text{otherwise.} \end{cases}$$

ι_C is exactly "the cost of being feasible": 0 inside C, infinite penalty outside. $N_C(x)$ collects all directions that point "outward" from C at x.

Convex Functions, the Subdifferential, and Fermat's Theorem VI

Running Example: The Normal Cone of the Unit Disk

At the **boundary** point $z = (0.6, 0.8) \in C$ (on the unit circle), the normal cone is the outward ray $N_C(z) = \{t\,(0.6, 0.8) : t \geq 0\}$: for any $y \in C$ (e.g. $y = (0, 0)$), $\langle y - z,\, t(0.6, 0.8)\rangle = t\langle(-0.6, -0.8), (0.6, 0.8)\rangle = -t(0.36 + 0.64) = -t \leq 0$ for $t \geq 0$, confirming the defining inequality. At an **interior** point such as $z = (0.3, 0.4)$ (norm $0.5 < 1$), every direction is "allowed," so $N_C(z) = \{0\}$.

Theorem 2.1 (Fermat's Theorem)

If $f : \mathcal{H} \to (-\infty, +\infty]$ is a proper convex mapping, then x^* is a **minimizer** of f *if and only if*

$$0 \in \partial f(x^*). \tag{2.7}$$

Condition (2.7) is also called the **first-order (necessary and sufficient) optimality condition**. It generalizes "set the derivative to zero" from calculus to arbitrary (possibly non-smooth) convex functions.

The Proximal Operator (Proximity Operator) I

Definition 2.18 (Proximity Operator)

Let $f \in \Gamma_0(\mathcal{H})$ and $\gamma > 0$ be a parameter. For any $x \in \mathcal{H}$, the **proximity operator** (or **proximal mapping**) of f parameterized by γ is defined by

$$\operatorname{prox}_{\gamma f}(x) := \operatorname*{argmin}_{y \in \mathcal{H}} \ f(y) + \frac{1}{2\gamma}\|y - x\|^2.$$

Plain-English Intuition

The proximal operator **nudges** x **toward minimizing** f**, without moving too far from** x: it trades off "obeying f" (making $f(y)$ small) against "staying close to x" (making $\|y - x\|^2$ small), with γ controlling the trade-off. Small γ favors staying close to x; large γ favors minimizing f more aggressively. Note that $\operatorname{prox}_{\gamma f}$ is always **firmly nonexpansive**.

The Proximal Operator (Proximity Operator) II

The Projection as a Special Case: $\operatorname{prox}_{\iota_C} = P_C$

Let C be a nonempty closed convex set and $f = \iota_C$. Then

$$\operatorname{prox}_{\gamma\iota_C}(x) = \operatorname*{argmin}_{y\in\mathcal{H}} \iota_C(y) + \frac{1}{2\gamma}\|y-x\|^2 = \operatorname*{argmin}_{y\in C} \frac{1}{2\gamma}\|y-x\|^2 = \operatorname*{argmin}_{y\in C} \|y-x\|^2 = P_C(x),$$

since $\iota_C(y) = +\infty$ rules out any $y \notin C$, leaving a plain nearest-point search inside C (the positive scalar $\frac{1}{2\gamma}$ does not change which y minimizes the expression). This holds for *every* $\gamma > 0$, so we simply write $\operatorname{prox}_{\iota_C} = P_C$.

Running Example: $\operatorname{prox}_{\iota_C}((3,4))$

Take $x = (3,4)$, $\gamma = 1$. Minimizing $\|y-(3,4)\|^2$ over $y \in C$ means finding the nearest point of the unit disk to $(3,4)$, which (since $\|x\| = 5 > 1$) is $y^* = x/\|x\| = (0.6, 0.8)$ – exactly $P_C(x)$ computed at the very start of this deck.

The Proximal Operator (Proximity Operator) III

The Link Back to Resolvents and Zeros

Combining (2.5), Definition 2.18, and Lemma 2.12: for $\gamma > 0$,

$$\operatorname{prox}_{\gamma f} = J_{\gamma \partial f}.$$

That is, the proximal mapping of f is exactly the resolvent of its (maximally monotone) subdifferential. Consequently, in terms of fixed points,

$$\operatorname{Fix}(\operatorname{prox}_{\gamma f}) = \operatorname{Fix}(J_{\gamma \partial f}) = \operatorname{zer}(\partial f).$$

In words: the points left unmoved by the proximal step are exactly the points where $0 \in \partial f(x)$, i.e. exactly the **minimizers** of f by Fermat's theorem. This is why proximal-point algorithms (which repeatedly apply $\operatorname{prox}_{\gamma f}$) converge to minimizers of f.

Python: Convex Analysis — Monotonicity of a Gradient

A basic sanity check: the gradient of a convex quadratic $f(x) = \frac{1}{2}x^\top Qx$ (with Q symmetric positive semidefinite) is a **monotone operator** in the sense of Definition 2.10, since $\langle x - y, \nabla f(x) - \nabla f(y)\rangle = \langle x - y, Q(x - y)\rangle \geq 0$ whenever $Q \succeq 0$.

```python
import numpy as np
rng = np.random.default_rng(2)

# f(x) = 0.5 * x^T Q x, Q symmetric positive semidefinite => grad f(x) = Q x
Q = np.array([[2.0, 0.3], [0.3, 1.0]])    # SPD => f convex
grad = lambda x: Q @ x

mono_ok = True
for _ in range(3000):
    x, y = rng.uniform(-5, 5, 2), rng.uniform(-5, 5, 2)
    if np.dot(x - y, grad(x) - grad(y)) < -1e-9:
        mono_ok = False

print("gradient of convex quadratic is monotone for all trials?", mono_ok)
```

Python: Convex Analysis — $\operatorname{prox}_{\iota_C} = P_C$ Numerically

We confirm $\operatorname{prox}_{\iota_C} = P_C$ by comparing the closed-form projection formula against a brute-force grid search minimizing $\|y - x\|^2$ over $y \in C$ (the definition of $\operatorname{prox}_{\iota_C}(x)$, since ι_C contributes 0 inside C and $+\infty$ outside).

```python
import numpy as np

def P_C(z):
    n = np.linalg.norm(z)
    return z if n <= 1.0 else z / n

def prox_indicator_bruteforce(x, n_grid=200):
    # brute-force argmin over C of ||y-x||^2 on a polar grid: r in [0,1]
    best_y, best_val = None, np.inf
    for r in np.linspace(0, 1, n_grid):
        for th in np.linspace(0, 2 * np.pi, n_grid, endpoint=False):
            y = r * np.array([np.cos(th), np.sin(th)])
            val = np.dot(y - x, y - x)
            if val < best_val:
                best_val, best_y = val, y
    return best_y

x_test = np.array([3.0, 4.0])
closed_form = P_C(x_test)
brute_force = prox_indicator_bruteforce(x_test)
print("closed form P_C(x)  =", closed_form)
print("brute-force prox(x) =", brute_force)
print("agree within grid resolution?",
```

Python: Confirming Both Figures

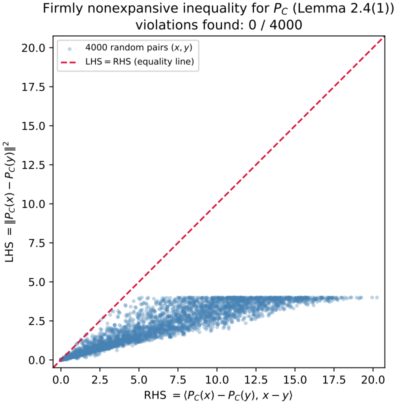
The two figures used in this deck were generated by `figures/gen_figures.py`: a picture of P_C acting on several points of $\mathbb{R}^2$, and a scatter plot confirming the firmly-nonexpansive inequality holds (LHS $\leq$ RHS) across thousands of random pairs.

```python
# Excerpt from figures/gen_figures.py (see that file for the full script)
import numpy as np

def project_to_unit_disk(x):
    x = np.asarray(x, dtype=float)
    norm = np.linalg.norm(x)
    return x.copy() if norm <= 1.0 else x / norm

n_samples = 4000
X = np.random.uniform(-4, 4, size=(n_samples, 2))
Y = np.random.uniform(-4, 4, size=(n_samples, 2))
lhs, rhs = np.empty(n_samples), np.empty(n_samples)
for i in range(n_samples):
    px, py = project_to_unit_disk(X[i]), project_to_unit_disk(Y[i])
    diff_p, diff_xy = px - py, X[i] - Y[i]
    lhs[i] = np.dot(diff_p, diff_p)      # ||P_C(x)-P_C(y)||^2
    rhs[i] = np.dot(diff_p, diff_xy)     # <P_C(x)-P_C(y), x-y>
violations = np.sum(lhs > rhs + 1e-9)
print(f"violations of LHS <= RHS: {violations} / {n_samples}")
# Output: violations of LHS <= RHS: 0 / 4000
```

The Firmly Nonexpansive Inequality: Numeric Confirmation



Every one of 4000 random pairs $(x, y) \in \mathbb{R}^2 \times \mathbb{R}^2$ satisfies $\|P_C(x) - P_C(y)\|^2 \leq \langle P_C(x) - P_C(y), x - y \rangle$: all points fall on or below the equality line.

Outline

1. Overview and the Running Example
2. Part (A): Notation
3. Part (B): Classes of Operators
4. Part (C): Convex Analysis
5. Chapter Summary

Chapter Summary: The Vocabulary Hierarchy I

Chapter Summary: The Vocabulary Hierarchy II

Operator Classes, From Strongest to Weakest

Class	**Key inequality / relation**
Contractive	$\|Tx - Ty\| \le L\|x - y\|$, $L < 1$
Firmly nonexpansive	$\|Tx - Ty\|^2 \le \langle Tx - Ty, x - y\rangle$
$\frac{1}{2}$-averaged	special case of averaged, $\alpha = \frac{1}{2}$
α-averaged	$T = \alpha S + (1-\alpha)\mathrm{Id}$, S nonexpansive
Nonexpansive	$\|Tx - Ty\| \le \|x - y\|$
Quasi-nonexpansive	nonexpansive inequality, $y \in \mathrm{Fix}(T)$ only
k-demicontractive	$\|Tx - y\|^2 \le \|x - y\|^2 + k\|Tx - x\|^2$
Pseudocontractive	$\|Tx - Ty\|^2 \le \|x - y\|^2 + \|(\mathrm{Id} - T)x - (\mathrm{Id} - T)y\|^2$

Each class in this list contains all classes above it (e.g. every firmly nonexpansive map is $\frac{1}{2}$-averaged, every averaged map is nonexpansive, every nonexpansive map with a fixed point is quasi-nonexpansive). P_C sits near the top: it is firmly nonexpansive, hence $\frac{1}{2}$-averaged, hence nonexpansive, hence (having fixed points) quasi-nonexpansive.

Chapter Summary: The Vocabulary Hierarchy III

Convex-Analysis Vocabulary, Side by Side

Object	**Role**
Monotone operator A	multivariate "increasing function"
Maximally monotone A	monotone, and cannot be enlarged
Resolvent $J_{\gamma A}$	$(\mathrm{Id} + \gamma A)^{-1}$, always firmly nonexpansive
β-cocoercive B	quantified monotone + Lipschitz
$f \in \Gamma_0(\mathcal{H})$	proper, convex, lower semi-continuous
∂f	subdifferential; maximally monotone if $f \in \Gamma_0(\mathcal{H})$
ι_C, $N_C = \partial \iota_C$	indicator function, normal cone
$\mathrm{prox}_{\gamma f}$	$J_{\gamma \partial f}$, firmly nonexpansive
prox_{ι_C}	equals P_C exactly

Chapter Summary: The Vocabulary Hierarchy IV

What Comes Next

Chapter 3 puts this vocabulary to work: the Krasnosel'skiĭ–Mann iteration $x_{n+1} = (1-\alpha_n)x_n + \alpha_n T(x_n)$ is precisely an averaging step applied to a nonexpansive (or more general) operator T, and its convergence proof will invoke Lemma 2.4 through 2.9, the demiclosedness principle, and the identity (2.4) introduced in this chapter – all without re-deriving them.